Credit Card Churn: Who Leaves, Why, and What to Do

Executive summary | Kaggle 'Credit Card Customers' dataset, 10,127 customers | All money in USD

16.1%	**63%**	**$30.6K**	**59%**
churn rate (1,627 of 10,127 customers)	of churners sit in the 'Disengaging' segment (23% of customers)	net benefit / year, base case (this 10,127-customer portfolio)	of churners reached by contacting only 24% of customers

What we found

• **Churn is a behavior problem.** Attrited customers made 45 transactions a year vs. 69 for existing ones (Cohen's d = -1.24, a large effect), and their transaction count fell sharply from Q1 to Q4. Gender, education, marital status, income and card tier are each weak predictors on their own (Cramer's V <= 0.04).

• **Warning signs are visible and measurable:** fewer products held, more months inactive and more service contacts each raise churn risk (chi-square, Holm-corrected p < 0.001).

• **Where to look:** the 'Disengaging' segment (2,285 customers) churns at 45%, 2.8x the average; single-product customers with no revolving balance churn at 34%.

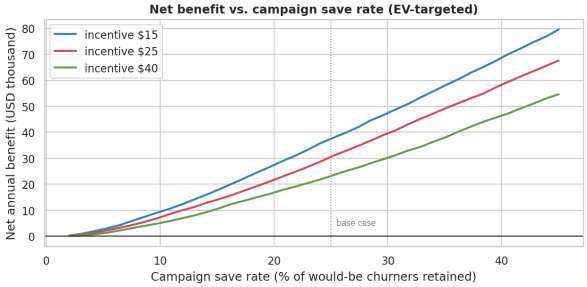
Recommended actions by segment

Segment	Customers	Churn	Action	Break-even save rate*	Net if contacted in full
S1 Disengaging	2,285	45%	Reactivation: spend-triggered cashback + early-warning alert	9%	$18,662
S2 Single-product, no revolving	380	34%	Relationship bundle: cross-sell 2nd product, fee/rate benefit	20%	$444
S3 Service friction	1,082	11%	Service recovery: senior callback, root-cause fix (selective)	31%	-$1,359
S4 High-value stable	1,954	5%	Priority care: relationship-manager review (selective)	32%	-$2,729

*Save rate at which the full-segment campaign breaks even (assumed rate: 25%). Blanket outreach loses money where churn is lower, so contact only customers with positive expected value.

What it is worth

Scoring every customer with the churn model and contacting only those whose expected value is positive means reaching **2,431 of 10,127 customers** (cost $12,192, expected saves ~241 customers). Base case: **$30,613 net per year** (ROI 251%). If only 10% of at-risk customers are retained: $7,142; at 35%: $49,091. Scaled linearly to a 1M-customer book (illustration): about **$3.0M per year** (range $0.7M to $4.8M).



Key assumptions (all editable in the notebook)

Revenue = 1.8% interchange on spend + 15% margin on revolving balance + annual fee. Contact cost $2; cashback $25, also claimed by 20% of non-churners. Save rate 25% (to be validated). One-year benefit only, measured on churners' current (already reduced) activity: conservative.

Next steps

1) A/B test: hold out ~10% of the target list as control to measure the true save rate.
2) Rebuild the score on a lagged window so alerts fire before customers disengage.
3) Replace the revenue proxy with real P&L data.